SIMPLE TYPES CUT OUT INITIAL PARTS THAT ARE TYPE-DEFINABLE, DISCRETELY ORDERED AND CLOSED UNDER MINIMA

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Allowed standard prerequisites (editorial).

- The definition of a simple type over A, witnessed by a subset C of the definable closure and a linear order (D,<). Set up in the definitions preceding the counted statement; used throughout the counted proof. Fixes the data the four clauses quantify over and the meaning of C-type used in each step.
- The uniqueness of the simple type p extending the C-type condition. Used inside the counted proof at line 155. Lets a point whose type is finitely satisfied in C but not algebraic be identified with p, giving the initial-part claim.

Lemma 0.1. *If $p \in S_1(A)$ is a simple type, as witnessed by $C \subset \mathrm{dcl}(A)$ and $(D, <)$, then:*

(a) $P(M) := C \cup p(M)$ is an initial part of $(D, <)$ that is type-definable over A.

(b) There exists an A-definable, discretely ordered, initial part $D_0 \subseteq D$ with $P(M) \subseteq D_0$. In particular, $(p(M), <)$ is a discrete order.

(c) Every nonempty, relatively M-definable subset of $P(M)$ has a minimum.

(d) If C and $(D_1, <_1)$ also witness the simplicity of p such that $<_1$ and $<$ agree on C, then they agree on $P(M)$, too.

Proof. To simplify the notation, assume $A = \varnothing$ and write $x < y$ instead of $x \in D \wedge y \in D \wedge x < y$.

(a) Since C is an initial part of $(D, <)$, $C < p(M)$ holds. To prove that $P(M)$ is an initial part of D, it suffices to show that $a \in p(M)$ and $b < a$ together imply $b \in P(M)$. Let $\phi(x) \in \mathrm{tp}(b)$. Then $\exists y(\phi(y) \wedge y < x) \in \mathrm{tp}(a) = p$. Since p is a C-type, there are $c \in C$ and $b' \in D$ with $\models \phi(b') \wedge b' < c$. Then $b' < c$ implies $b' \in C$, so $\phi(x)$ is realized by $b' \in C$. We have just shown that every formula $\phi(x) \in \mathrm{tp}(b)$ is realized in C, so the type $\mathrm{tp}(b)$ is finitely satisfied in C; If $\mathrm{tp}(b)$ is algebraic then $b \in C$; otherwise, $\mathrm{tp}(b)$ is

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a C-type, so $\mathrm{tp}(b) = p$ holds by the uniqueness of p. In both cases we have $b \in P(M)$, so $P(M)$ is an initial part of D. Then it is is easy to see that $P(M)$ is defined by $P(x) := \{\exists y(\theta(y) \wedge x < y) \mid \theta(y) \in p(y)\}$.

(b) Let $c_0 = \min C$ and let $\phi(x)$ be a formula expressing that the interval $[c_0, x]$ is discretely ordered by $<$. For all $c \in C$ the interval $[c_0, c]$ is finite, so $\models \phi(c)$ holds. Since p is a C-type, $\phi(x) \in p$. Let D_0 be the union of all intervals $[c_0, a] \subset D$ where $a \in \phi(M)$.

(c) Let $\varnothing \neq D_1 \subset P(M)$ where D_1 is relatively defined by $\theta(x, \bar{b})$ ($\bar{b} \in M$). Pick $a \in p(M)$ with $[c_0, a] \cap D_1 \neq \varnothing$. Let $\phi(x, \bar{y})$ be a formula expressing that the set $[c_0, x] \cap \theta(M, \bar{y})$ is nonempty and has minimum. In particular, $\models \phi(a, \bar{b})$ holds. For all $c \in C$ the interval $[c_0, c]$ is finite, so $\models \forall \bar{y}\,(\exists z(c_0 \leqslant z \leqslant c \wedge \theta(z, \bar{y})) \to \phi(c, \bar{y}))$. Since p is a C-type and $a \in p(M)$,

$$\models \forall \bar{y}\,(\exists z(c_0 \leqslant z \leqslant a \wedge \theta(z, \bar{y})) \to \phi(a, \bar{y})).$$

In particular, $\models \exists z(c_0 \leqslant z \leqslant a \wedge \theta(z, \bar{b})) \to \phi(a, \bar{b})$. Now $[c_0, a] \cap D_1 \neq \varnothing$ implies $\models \exists z(c_0 \leqslant z \leqslant a \wedge \theta(z, \bar{b}))$, so $\models \phi(a, \bar{b})$ and $\min([c_0, a] \cap D_1)$ exists. Hence $\min D_1$ exists, too.

(d) Suppose not. Then for all $a \in p(M)$ the set $D(a)$ defined by $x < a \wedge x >_1 a$ is nonempty so, by part (c), $b = \min_< D(a)$ is well-defined. By part (a), $C <_1 p(M)$ implies $b \notin C$, so $b < a$ implies $b \in p(M)$. In particular, $D(b) \neq \varnothing$ and for any $b' \in D(b)$ we have $b' < b < a \wedge b' >_1 b >_1 a$, contradicting the minimality of b in $D(a)$. □

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